

100 Days of Linux - Day 029

Title: Changing Permissions with chmod

Today's Goal

Learn how to change read, write, and execute permissions using the **chmod** command.

Why This Matters

Developers and data engineers regularly make shell scripts executable, protect configuration files, and control access to project resources.

Command of the Day: chmod

Symbolic Syntax: chmod u+x script.sh

chmod g-w notes.txt

chmod o+r report.txt

Numeric Syntax: chmod 755 script.sh

chmod 644 notes.txt

Meaning:

u = User (Owner)

g = Group

o = Others

a = All

r = Read (4)

w = Write (2)

x = Execute (1)

Beginner Lesson

Yesterday you learned how to read permissions using **ls -l**. Today you'll change them with **chmod**. Start with symbolic mode because it's easier to understand. Then try numeric mode, which is widely used in real projects.

Practice Questions

1. Create a file named script.sh and view its permissions using **ls -l**.
2. Give the owner execute permission on script.sh using symbolic mode, then verify the change.
3. Create a file named report.txt and remove write permission from the group.
4. Change the permissions of report.txt to 644 using numeric mode and verify with **ls -l**.
5. Display your current working directory after completing today's tasks.

Hints

Always check permissions before and after using chmod by running **ls -l.**

Mini Challenge

Create files app.py, pipeline.sh, and config.txt. Make pipeline.sh executable (755), keep config.txt read/write for the owner only

(600), and compare the permissions with `ls -l`.

Common Mistakes

Forgetting to verify permission changes or confusing symbolic mode with numeric mode.

Did You Know?

A shell script usually needs execute (x) permission before it can be run directly from the terminal.

Pro Tip

For regular text files, 644 is a common permission. For executable scripts, 755 is a common choice.

Answer Key (Instructor Only)

Q1: `touch script.sh && ls -l`

Q2: `chmod u+x script.sh && ls -l script.sh`

Q3: `touch report.txt && chmod g-w report.txt && ls -l report.txt`

Q4: `chmod 644 report.txt && ls -l report.txt`

Q5: `pwd`

Homework

Create five files and practice changing their permissions using both symbolic and numeric `chmod`.

Next Day Preview

Tomorrow you'll learn common Linux permission values (644, 755, 600, 700) and when to use each in real-world projects.